

AMEER SALMAN M

Data Engineer — regulated payments data, streaming pipelines, reconciliation at scale

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SUMMARY

Data engineer with close to seven years at Amazon Pay building and operating the pipelines behind India's regulated payment reconciliation, exiting as Data Engineer II. Rebuilt batch reporting as event-driven streaming, cutting downstream SLA from 36–40 hours to 30 minutes with eight months of zero-defect production running. Reconciled roughly 400 million transactions a month across ~15 source systems and ~120 ETL jobs, under a statutory five-working-day reporting deadline. Comfortable owning a system end to end — including the compliance constraints, the finance business rules, and the handover that lets somebody else run it.

EXPERIENCE

Data Engineer II · Amazon Pay

OCT 2017 — JUL 2024

BENGALURU & CHENNAI | 6 YRS 10 MOS | EXIT TITLE

- Rebuilt batch order-transaction reporting as event-driven streaming ingestion using database change streams, cross-account S3 replication and Lambda — cutting downstream SLA from **36–40 hours to 30 minutes** (~99%), live **8 months with zero defects**.
- Built a near real-time merchant metrics pipeline across accounts and regions with customer-managed KMS encryption, partitioned Parquet, Glue crawlers, Athena and Redshift — **15–30 minute SLA, 100% accuracy, zero job failures**, with restricted PII excluded in flight before durable write.
- Automated a **20 million record** negative-remittance load with Python and Boto3 including pre-commit reconciliation — 100% accuracy, ~2 days of manual effort removed, clearing a multi-billion-rupee exception backlog.
- Orchestrated **~60 report jobs and 30 dashboards** into three one-click workflows and automated variance root-cause analysis (~4 days → automated), holding the **5-working-day** regulatory substantiation deadline and moving substantiation from monthly to daily.
- Introduced pipeline-as-code CI/CD across two data platforms with **dual sign-off SQL review** (business logic and query efficiency), eliminating drift between deployed jobs and source control.
- Built a self-mutating **AWS CDK** deployment pipeline spanning Beta, Gamma and Prod across multiple AWS accounts and regions, with approval workflow, security scanning, integration tests and bake time.
- Ran reconciliation across **~15 source systems and ~120 ETL jobs**, categorising **~400 million transactions/month** into daily and monthly transaction-level reports; reduced manual ETL workload 60%.
- Diagnosed and fixed silent exclusion of undocumented transaction types in the bank statement parser (Java) that was breaking closing-balance reconciliation — **50% less manual exception clearing**.
- Led migration of Redshift, QuickSight and S3 to the Mumbai region for **RBI payment aggregator data localisation**; supported RBI audit root-cause analysis across India regulatory accounts.
- Reclaimed **20% of Redshift disk** and eliminated CPU-driven cluster outages by replacing cluster-wide vacuum with targeted ANALYZE and deep-copy on high-impact tables.
- Delivered a cross-border EU pipeline under **data residency restrictions** using encrypted S3 replication, Lambda and Glue — 15–30 minute SLA, eliminating a recurring Sev-3 ticket.
- As Automation Specialist, built Python and Selenium automation for campaign management and seller onboarding, saving **~15 FTE** of effort; authored SOPs enabling stakeholder self-service.

TECHNICAL SKILLS

LANGUAGES	Python, SQL, Java, Bash	INGEST	DynamoDB Streams, Kinesis, S3 replication, Lambda, event triggers
PROCESSING	AWS Glue, EMR, PySpark, Pandas, Parquet	WAREHOUSE	Redshift, Athena, MySQL, DynamoDB
PLATFORM	AWS CDK, CloudFormation, Managed Airflow, CI/CD, Pytest	REPORTING	QuickSight, dashboard automation via Boto3 / AWS CLI
SECURITY	IAM, KMS (customer-managed keys), CloudWatch, secrets management	GOVERNANCE	GDPR in a data lake, PII exclusion, EU data residency, RBI localisation
DOMAIN	Payments reconciliation, financial substantiation, exception management	PRACTICES	Code review, SOP authoring, stakeholder sign-off, on-call

SELECTED PROJECTS

- **AWS automation toolkit** — Python tooling for EC2 snapshot lifecycle management. github.com/ameerafi
- **Meteorite proximity finder** — Python service over NASA open data, ranking landing sites by distance.

EDUCATION & CERTIFICATIONS

DEGREE	B.E. Mechanical Engineering — Park College of Engineering & Technology, Coimbatore, 2017	CERT	Add certifications
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